

TESTING – TEST PLANS AND TEST CASES

Aim:

Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Procedure

1. Understand Core Features of the Application

- User Registration & Login
 - Email Verification and Authentication
 - Entering Symptoms and Health Details
 - Uploading Medical Reports
 - Predicting Diseases using ML Model
 - Viewing Prediction Results and Severity
 - Receiving Health Recommendations
 - Viewing Prediction History
 - Admin Monitoring and Model Improvement
-

2. Define User Interactions

- Each test case simulates real user behavior in the Smart Health Prediction System.
 - Example interactions:
 - User registers with valid details
 - User uploads a medical report
 - User enters symptoms for prediction
 - Admin monitors system performance
 - User views previous health predictions
-

3. Design Happy Path Test Cases

- Focused on validating that all modules work correctly under normal conditions.
 - Example scenarios:
 - User successfully registers and verifies email
 - User logs in successfully
 - Symptoms are submitted successfully
 - Disease prediction result is generated correctly
 - Recommendation tips are displayed properly
 - Prediction history loads successfully
-

4. Design Error Path Test Cases

- Simulate invalid or unexpected conditions to verify system robustness and error handling.
 - Example scenarios:
 - Registration fails with duplicate email
 - Weak password validation error
 - Login fails with incorrect credentials
 - Invalid symptom input accepted
 - Medical report upload failure
 - ML prediction API timeout
 - Prediction history not loading
 - Email verification token expired
-

5. Break Down Steps and Expected Results

- Each test case includes:
 - Step-by-step user actions
 - Input conditions
 - Expected system response/output
- Ensures clarity for:
 - Manual testing
 - Automation testing
 - Azure Test Plans execution

Example:

- Step: User enters symptoms and clicks Predict
 - Expected Result: System displays predicted disease with confidence score and recommendations
-

6. Use Clear Naming and IDs

- Test cases are identified using unique IDs and descriptive names.

Examples:

- TC01 – Successful User Registration
- TC02 – Invalid Login Attempt
- TC03 – Upload Medical Report Successfully
- TC04 – Disease Prediction Generated
- TC05 – Prediction History Display
- TC06 – Expired Email Verification Link
- Helps in:
 - Easy tracking
 - Mapping with user stories
 - Bug reporting in Azure DevOps

7. Separate Test Suites

- Test cases are grouped according to modules/features.

Example Test Suites:

- Authentication Test Suite
 - Registration
 - Login
 - Email Verification
- Health Data Test Suite
 - Symptom Input
 - Medical Report Upload
- Prediction System Test Suite
 - Disease Prediction
 - Result Display
 - Recommendation System
- Admin Test Suite
 - System Monitoring
 - ML Model Improvement
- Non-Functional Test Suite
 - Performance Testing
 - Security Testing
 - Reliability Testing
- Improves organization and execution flow in Azure DevOps.

8. Prioritize and Review

- Critical functionalities are marked as high priority.

High-Priority Features:

- User Authentication
- Disease Prediction Accuracy
- Secure Medical Data Storage
- Email Verification
- System Availability
- Regular review and bug tracking ensure:
 - Better reliability
 - Improved prediction accuracy
 - Stable cloud deployment on Microsoft Azure



• New test plan

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Smart Health Prediction...

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Overview

Boards

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Parameters

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New Test Plan

Name *

Enter a plan name

Area Path *

Smart Health Prediction System

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Iteration *

Smart Health Prediction System

▼

Create

Cancel

• Test suite

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Test Suites

Filter suites by name

▼ Smart health prediction system Test...

Click login (2)

New Suite

Enter valid email (1)

Click login (ID: 46)

Define

Execute

Chart

Test Points (2 items)

▼

Run for web application

▼

Title	Outcome	Order	Test Case Id
enter mail id	Active	1	47
Test	Active	2	49

1. Test case

Give two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

USER STORIES

- As a user, I want to register and log in securely so that my health data is protected (ID: 79).
 - As a user, I want to enter my symptoms so that the system can analyze my health condition (ID: 76).
 - As a user, I want to upload medical reports so that prediction accuracy improves (ID: 73).
 - As a user, I want to receive disease predictions so that I can take precautions early (ID: 68).
 - As a user, I want to view recommendations and prediction history so that I can track my health status (ID: 65).
-

Test Suit: TS01 - User Registration & Login (ID: 86)

1. TC01 – Successful Registration

- Action:
 - Go to the Registration page.
 - Enter valid username, email, and password.
 - Click "Register".
- Expected Results:
 - Registration form accepts values.
 - Account is created successfully.
 - User receives verification email.
- Type: Happy Path

2. TC02 – Login with Invalid Password

- Action:
 - Go to Login page.
 - Enter valid email and wrong password.
 - Click "Login".
 - Expected Results:
 - Error message "Invalid credentials" is displayed.
 - Type: Error Path
-

Test Suit: TS02 - Enter Symptoms (ID: 87)

1. TC03 – Enter Symptoms Successfully

- Action:
 - Login successfully.
 - Open symptom input page.
 - Enter valid symptoms.
 - Click "Submit".
- Expected Results:
 - Symptoms are stored successfully.
 - User proceeds to prediction page.
- Type: Happy Path

2. TC04 – Invalid Symptom Submission

- Action:
 - Open symptom input page.
 - Leave fields empty.
 - Click "Submit".
 - Expected Results:
 - Error message "Please enter symptoms" is shown.
 - Type: Error Path
-

Test Suit: TS03 - Upload Medical Reports (ID: 88)

1. TC05 – Upload Report Successfully

- Action:
 - Login successfully.
 - Select a valid medical report file.
 - Click "Upload".
- Expected Results:
 - File uploads successfully.
 - Confirmation message is displayed.
- Type: Happy Path

2. TC06 – Upload Unsupported File Format

- Action:
 - Select unsupported file type.
 - Click "Upload".
 - Expected Results:
 - Error message "Invalid file format" is displayed.
 - Type: Error Path
-

Test Suit: TS04 - Disease Prediction (ID: 89)

1. TC07 – Generate Disease Prediction

- Action:
 - Enter symptoms.
 - Click "Predict".
- Expected Results:
 - Predicted disease with confidence score is displayed.
- Type: Happy Path

2. TC08 – Prediction API Failure

- Action:
 - Enter symptoms.
 - Disconnect network or stop API service.
 - Click "Predict".
 - Expected Results:
 - Error message "Prediction service unavailable" is displayed.
 - Type: Error Path
-

Test Suit: TS05 - Recommendations & Prediction History (ID: 90)

1. TC09 – View Recommendation and History

- Action:
 - Login successfully.
 - Open prediction history page.
- Expected Results:
 - Previous predictions and recommendations are displayed correctly.
- Type: Happy Path

2. TC10 – History Loading Failure

- Action:
 - Disconnect internet connection.
 - Open prediction history page.
- Expected Results:
 - Error message "Unable to load history" is displayed.
- Type: Error Path

Test Case

chrome web store

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 **Test & Feedback**

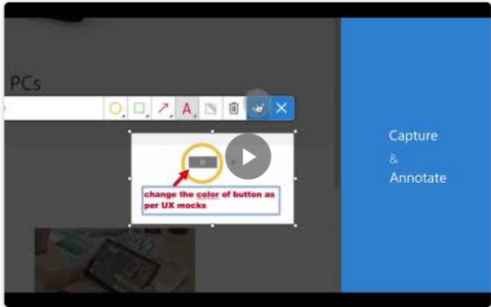
Add to Chrome

4.2★ (176 ratings) ⓘ < Share

Extension

Workflow & Planning

200,000 users



 NEW TEST CASE * Field 'Title' cannot be empty.

Enter title

 jeevanR

0 Comments

Add Tag

State

Design

Area

smart health prediction system

Reason

New

Iteration

smart health prediction system\Iteration 1

Steps

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Steps	Action		Expected result	

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Test Suites

Filter suites by name

Smart health prediction system Test...

Click login (2)

New Suite

Enter valid email (1)

Enter valid email (ID: 44)

Define

Execute

Chart

Test Points (1 item)

Title

Successful Login

Outcome

Order

Test Case Id

Active

1

48

Extensions

Full access

These extensions can see and change information on this site.

Test & Feedback

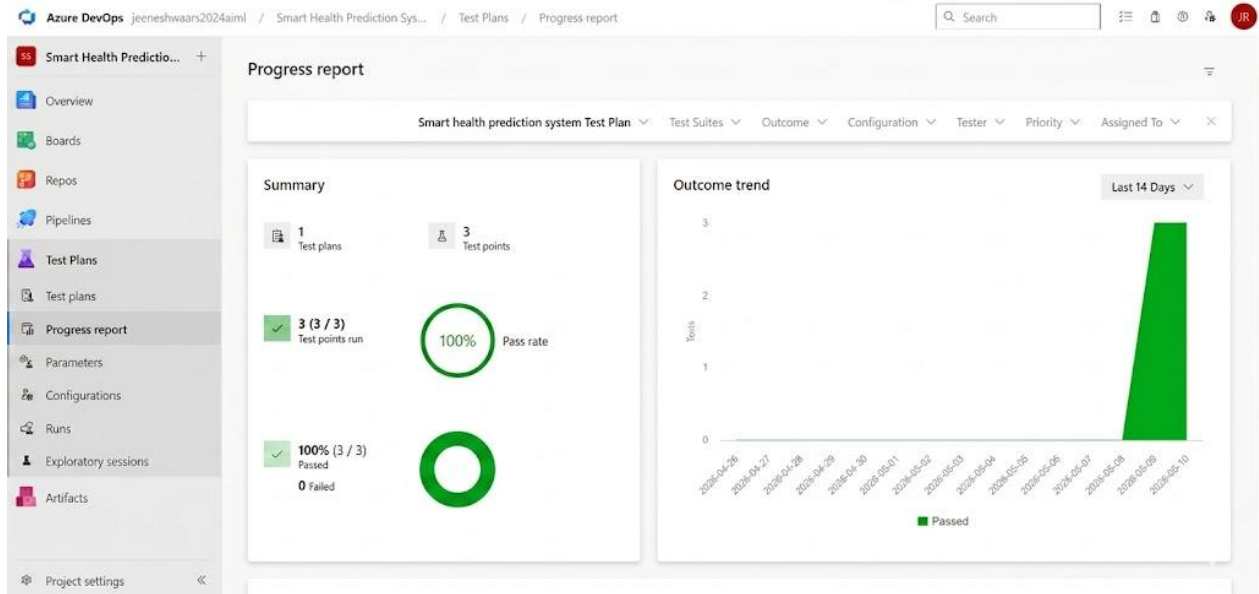
Manage extensions

Search

Help

Run for web application

1. Progress report



Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path